Package ‘breaktest’

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Title Set of unit root and cointegration tests

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Depends R (>= 3.5.0)

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Imports doSNOW, foreach, stringr, Rfast

Description This package contains a set of different unit root and cointegration tests in the presence of structural breaks in the data.

License GPL (>= 2)

URL https://github.com/d9d6ka/breaktest

BugReports https://github.com/d9d6ka/breaktest/issues

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Contents

`breaktest.KP` *Kejrival-Perron procedure of breaks number detection*

Description

Kejrival-Perron procedure of breaks number detection

Usage

```
breaktest.KP(y, const = FALSE, breaks = 1, criterion = "bic", trim = 0.15)
```

Arguments

`y`	An input series of interest.
`const`	Whether the break in constant is allowed.
`breaks`	Number of breaks.
`criterion`	Needed information criterion: aic, bic, hq or lwz.
`trim`	A trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

The estimated optimal break point.

References

Kejriwal, Mohitosh, and Pierre Perron. “A Sequential Procedure to Determine the Number of Breaks in Trend with an Integrated or Stationary Noise Component: Determination of Number of Breaks in Trend.” Journal of Time Series Analysis 31, no. 5 (September 2010): 305–28. https://doi.org/10.1111/j.1467-9892.2010.00666.x.

coint.conf.sets *Confidence sets for the break date in cointegrating regressions*

Description

This procedure is to construct a confidence set for the change point in cointegrating regressions.

Usage

```
coint.conf.sets(
  y,
  trend = FALSE,
  zb = NULL,
  zf = NULL,
  nF = NULL,
  nL = NULL,
  conf.level = 0.9,
  trim = 0.05,
  criterion = "bic"
)
```

Arguments

`y`	A time series of interest.
`trend`	Whether the trend is to be included.
`zb`	I(1) regressors with break.
`zf`	I(1) regressors without break.
`nF`, `nL`	Number of leads and lags of `z` regressors. If any is NULL then both are estimated using informational `criterion`.
`conf.level`	Confidence level to obtain appropriate critical values.
`trim`	The trimming parameter to find the lower and upper bounds of possible break date.
`criterion`	A criterion for lead and lag number estimation.

Value

A list of confidence sets.

References

Kurozumi, Eiji, and Anton Skrobotov. “Confidence Sets for the Break Date in Cointegrating Regressions.” Oxford Bulletin of Economics and Statistics 80, no. 3 (2018): 514–35. https://doi.org/10.1111/obes.12223.

coint.CSS *KPSS-based cointegration test with multiple known structural breaks*

Description

Procedure to test the presence of cointegration in the case of multiple known structural breaks using KPSS test. This procedure is based on the procedures for one and two known breaks by Carrion-i-Silvestre and Sansó (2006, 2007).

Usage

```
coint.CSS(
  y,
  x = NULL,
  const = FALSE,
  trend = FALSE,
  break.type,
  break.point,
  break.coint = FALSE,
  weakly.exog = TRUE,
  max.lags,
  max.leads,
  criterion = "bic",
  lr.kernel = "Kurozumi",
  lr.lag = NULL,
  boot.p = FALSE,
  boot.type = "sample",
  boot.iter = 999
)
```

Arguments

`y`	A time series of interest.
`x`	A matrix of explanatory stochastic regressors. If it's NULL then the result of this function is just a KPSS test for y.
`const, trend`	Whether a constant or trend should be included.
`break.type`	A single value or vector of • `c`: for the break in const, • `t`: for the break in trend, • `ct`: for the break in const and trend.
`break.point`	Array of structural break moments.
`break.coint`	Whether breaks in cointegrating relation are needed to be included. See Carrion-i-Silvestre & Sansó (2006) for details.
`weakly.exog`	Boolean where we specify whether the stochastic regressors are exogenous or not • `TRUE`: if the regressors are weakly exogenous, • `FALSE`: if the regressors are not weakly exogenous (DOLS is used in this case).

`max.lags`, `max.leads`
: Scalars defininig the initial number of lags and leads for DOLS.

`criterion`
: Information criterion for DOLS lags and leads selection: aic, bic, hq, or lwz,

`lr.kernel`
: Kernel for calculating long-run variance

- `Kurozumi` for the Kurozumi's proposal, using Quadratic kernel (default).
- `Bartlett`: for Bartlett kernel,
- `Quadratic`: for Quadratic Spectral kernel,

`lr.lag`
: scalar showing the bandwidth of long run variance estimator. If negative or `NULL` then the bandwidth is selected as in Andrews (1991).

`boot.p`
: Whether bootstrapped p-values should be returned.

`boot.type`
: Method of building synthetic y:

- `sample` — using a sample from residuals,
- `Cavaliere-Taylor` — multiplying residuals by $N(0, 1)$,
- `Rademacher` — multiplying residuals by ± 1.

`boot.iter`
: The number of bootstrap iterations,

`...`
: A dummy parameter for technical purposes. Just ignore it.

If you want to get results like in Carrion-i-Silvestre & Sansó (2006) then you should

- provide a `break.point` of length 1;
- for the specification of `An` set `const=TRUE`, `trend=FALSE`, `break.type=c("c")`;
- for the specification of `A` set `const=TRUE`, `trend=TRUE`, `break.type=c("c")`;
- for the specification of `B` set `const=TRUE`, `trend=TRUE`, `break.type=c("t")`;
- for the specification of `C` set `const=TRUE`, `trend=TRUE`, `break.type=c("ct")`;
- for the specification of `D` set `const=TRUE`, `trend=FALSE`, `break.type=c("c")`;
- for the specification of `E` set `const=TRUE`, `trend=TRUE`, `break.type=c("ct")`, and `break.coint=TRUE`.

If you want to get results like in Carrion-i-Silvestre & Sansó (2007) then you should

- provide a `break.point` of length 2;
- for the specification of `AAn` set `const=TRUE`, `trend=FALSE`, `break.type=c("c", "c")`;
- for the specification of `AA` set `const=TRUE`, `trend=TRUE`, `break.type=c("c", "c")`;
- for the specification of `BB` set `const=TRUE`, `trend=TRUE`, `break.type=c("t", "t")`;
- for the specification of `CC` set `const=TRUE`, `trend=TRUE`, `break.type=c("ct", "ct")`;
- for the specification of `AB-BA` set `const=TRUE`, `trend=FALSE`, `break.type=c("c", "t")`;
- for the specification of `AC-CA` set `const=TRUE`, `trend=TRUE`, `break.type=c("c", "ct")`;
- for the specification of `BC-CB` set `const=TRUE`, `trend=TRUE`, `break.type=c("t", "ct")`.

Details

At the first step obtained are the residuals from the regression of LHS series y on the set of deterministic components and (if any) exogenous regressors x. Deterministic components may include

- constant term,
- trend component,
- constant with break $DU_{t,\tau} = 1[t > \tau]$,
- trens with break $DT_{t,\tau} = 1[t > \tau](t - \tau)$.

If x are weakly exogenous then the regression is estimated by OLS, if not then DOLS regression is applied. The procedure also allows for the break in the cointegrating equation, in that case products of $DU_{t,\tau}$ and x are added to the regressors list.

After the residuals are obtained KPSS test statistic is calculated using `kernel`. If it's not set then QS kernel is used with banwidth selection as in Andrews (1991), and with Kurozumi (2002) proposal of banwidth limiting.

p-value (if needed) is calculated by bootstrapping procedure.

Value

An object of class `bt_kpss` containing

- `statistic`: the value of test statistic,
- `coefficients`: OLS/DOLS estimates of the coefficients,
- `se.coefs`: standard errors of the coefficients above,
- `t.stats`: t-statistics for the coefficients above,
- `residuals`: Residuals of the model,
- `t.beta`: t-statistics for `beta`,
- `fitted.values`: fitted values of the estimated model,
- `endog`: final y vector used in the estimated model,
- `exog`: final x matrix used in the estimated model,
- `DOLS.lags`: The estimated number of lags and leads in DOLS,
- `break.type`, `break.point`, `break.coint`: breaks specification,
- `criterions`: values of the information criterions for the estimated model,
- `lags`, `leads`: number of lags and leads in the estimated model.

References

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "Testing the Null of Cointegration with Structural Breaks." Oxford Bulletin of Economics and Statistics 68, no. 5 (October 2006): 623–46. https://doi.org/10.1111/j.1468-0084.2006.00180.x.

Carrion-i-Silvestre, Josep Lluís, and Andreu Sansó. "The KPSS Test with Two Structural Breaks." Spanish Economic Review 9, no. 2 (May 16, 2007): 105–27. https://doi.org/10.1007/s10108-006-9017-8.

#' Andrews, Donald W. K. "Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation." Econometrica 59, no. 3 (1991): 817–58. https://doi.org/10.2307/2938229.

Kurozumi, Eiji. "Testing for Stationarity with a Break." Journal of Econometrics 108, no. 1 (May 1, 2002): 63–99. https://doi.org/10.1016/S0304-4076(01)00106-3.

coint.GH *Gregory-Hansen test for the absense of cointegration*

Description

Gregory and Hansen (1996) test for the null hypothesis of no cointegration under a possible structural break at the unknown moment of time.

Usage

```
coint.GH(
  ...,
  shift = "level",
  trim = 0.15,
  max.lag = 10,
  criterion = "aic",
  add.cvals = TRUE
)
```

Arguments

`...`	Variables of interest.
`shift`	Expected break type.
`trim`	The trimming parameter to calculate break moment bounds.
`max.lag`	The maximum number of lags for the internal ADF testing.
`criterion`	The criterion for lag selection.
`add.cvals`	Whether critical values are to be returned. This argument is needed to suppress the calculation of critical values during the precalculation of tables needed for the p-values estimating.

Details

Gregory and Hansen (1996) test for the null hypothesis of no cointegration under a possible structural break at the unknown moment of time.

The authors proposed ADF- and Z-type tests, slightly modified to allow the presence of a possible regime shift. Three type of shifts are allowed:

- a shift in the constant,
- a shift in the constand with the trend included,
- and a shift in the constant and the cointegrating vector.

Critical values are calculated via the adopted MacKinnon procedure of estimating the model for the response surface.

Value

An object of type `cointGH`. It's a list of

- `shift`: shift type,
- `Za`: MZ_α statistic and c.v.,
- `Zt`: MZ_t statistic and c.v.,
- `ADF`: ADF statistic and c.v..

References

MacKinnon, James G. “Critical Values for Cointegration Tests.” Working Paper. Working Paper. Economics Department, Queen’s University, January 2010. https://ideas.repec.org/p/qed/wpaper/1227.html.

Gregory, Allan W., and Bruce E. Hansen. “Residual-Based Tests for Cointegration in Models with Regime Shifts.” Journal of Econometrics 70, no. 1 (January 1, 1996): 99–126. https://doi.org/10.1016/0304-4076(69)41685-7.

`coint.PR` *A set of residual based tests for cointegration*

Description

A set of residual based tests for cointegration

Usage

```
coint.PR(y, x, deter, kmin = 0, signif = 0.05)
```

Arguments

`y, x`	Variables of interest. `x` can be a matrix of several variables.
`deter`	A value equal to • 1: quasi-demeaned y and x, • 2: quasi-detrended y and x, • 3: quasi-demeaned y and quasi-detrended x.
`kmin`	A minimum number of lags to be used in the tests.

Details

The code provided is the original GAUSS code by Perron and Rodríguez ported to R.

Value

A list of:

- 7x1-matrix of test statistics values,
- estimated number of lags.

References

Perron, Pierre, and Gabriel Rodríguez. “Residuals-based Tests for Cointegration with Generalized Least-squares Detrended Data.” The Econometrics Journal 19, no. 1 (February 1, 2016): 84–111. https://doi.org/10.1111/ectj.12056.

`coint.VECM` *Test of the cointegration rank with a possible break in trend*

Description

This procedure is aimed on the problem of testing for the cointegration rank of a vector autoregressive process in the case where a trend break may potentially be present in the data.

Usage

```
coint.VECM(y, r, max.lag, trim = 0.15)
```

Arguments

`y`	A matrix of n VAR variables.
`r`	The cointegration rank tested against the alternative of n.
`max.lag`	The maximum number of lags.
`trim`	The trimming parameter to find the lower and upper bounds of possible break dates.

Details

This procedure is aimed on the problem of testing for the cointegration rank of a vector autoregressive process in the case where a trend break may potentially be present in the data.

The test is based on estimating the quasi log likelihood for two situations, with break, and without it. The one with the smallest value is considered to be the result.

The code provided is the original GAUSS code by Harris et al. ported to R.

Value

A list of:

- the indicator of the rejection of null.
- the estimated break point.
- the estimated lag number.

References

Harris, David, Stephen J. Leybourne, and A. M. Robert Taylor. "Tests of the Co-Integration Rank in VAR Models in the Presence of a Possible Break in Trend at an Unknown Point." Journal of Econometrics, Innovations in Multiple Time Series Analysis, 192, no. 2 (June 1, 2016): 451–67. https://doi.org/10.1016/j.jeconom.2016.02.010.

`info.criterions` *Information criterions*

Description

Information criterions

Usage

```
info.criterions(resids, extra, modification = FALSE, alpha = 0, y = NULL)
```

Arguments

`resids`	Input residuals needed for estimating the values of information criterions.
`extra`	Number of extra parameters needed for estimating the punishment term.
`modification`	Whether the unit-root test modificaton is needed. See Ng and Perron (2001) for further information.
`alpha`	The coefficient α of y_{t-1} in ADF model. Needed only for criterion modification purposes.
`y`	The vector of y_{t-1} in ADF model. Needed only for criterion modification purposes.

Details

Calculating the value of the following informational criterions:

- Akaike,
- Schwarz (Bayesian),
- Hannan-Quinn,
- Liu et al.

Value

The list of information criterions values.

References

Ng, Serena, and Pierre Perron. "Lag Length Selection and the Construction of Unit Root Tests with Good Size and Power." Econometrica 69, no. 6 (2001): 1519–54. https://doi.org/10.1111/1468-0262.00256.

`KP.seq.statistic` *Sequential statistic for breaks at unknown date.*

Description

Sequential statistic for breaks at unknown date.

Usage

```
KP.seq.statistic(
  y,
  const = FALSE,
  breaks = 1,
  criterion = "aic",
  trim = 0.15,
  max.lag = trunc(12 * (length(y)/100)^(1/4))
)
```

Arguments

`y`	A time series of interest.
`const`	Allowing the break in constant.
`breaks`	A number of breaks.
`criterion`	Needed information criterion: aic, bic, hq or lwz.
`trim`	A trimming value for a possible break date bounds.
`max.lag`	The maximum possible lag in the model.

Details

This procedure is based on ideas of Perron & Yabu (2009).

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

An estimated Wald statistic.

References

Kejriwal, Mohitosh, and Pierre Perron. "A Sequential Procedure to Determine the Number of Breaks in Trend with an Integrated or Stationary Noise Component: Determination of Number of Breaks in Trend." Journal of Time Series Analysis 31, no. 5 (September 2010): 305–28. https://doi.org/10.1111/j.1467-9892.2010.00666.x.

`MDF.1br` *MDF procedure for a single unknown break.*

Description

MDF procedure for a single unknown break.

Usage

```
MDF.1br(y, const = FALSE, trend = FALSE, trim = 0.15, criterion = "bic")
```

Arguments

`y`	A time series of interest.
`const`	Whether the constant term should be included.
`trend`	Whether the trend term should be included.
`trim`	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of sublists each containing

- The value of statistic: $MDF-GLS$, $MDF-OLS$,
- The asymptotic critical values. UR values are included as well.

`MDF.mlt` *MDF procedure for multiple unknown breaks.*

Description

MDF procedure for multiple unknown breaks.

Usage

```
MDF.mlt(y, const = FALSE, breaks = 1, breaks.star = 1, trim = 0.15, ZA = FALSE)
```

Arguments

`y`	A time series of interest.
`const`	Whether the constant term should be included.
`breaks`	Number of breaks.
`breaks.star`	Number of breaks got from the Kejrival-Perron procedure.
`trim`	Trimming value for a possible break date bounds.
`ZA`	Whether ZA variant should be used.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of sublists each containing

- The value of statistic: $MDF - GLS$, $MDF - OLS$,
- The asymptotic critical values. UR values are included as well.

PY.statistic *Perron-Yabu (2009) statistic for break at unknown date.*

Description

Perron-Yabu (2009) statistic for break at unknown date.

Usage

```
PY.statistic(
  y,
  const = FALSE,
  trend = FALSE,
  criterion = "aic",
  trim = 0.15,
  max.lag = trunc(12 * (length(y)/100)^(1/4))
)
```

Arguments

y	A time series of interest.
const, trend	Allowing the break in constant or trend.
criterion	Needed information criterion: aic, bic, hq or lwz.
trim	A trimming parameter to determine the lower and upper bounds for a possible break point.
max.lag	The maximum possible lag in the model.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

Value

A list of the estimated Wald statistic as well as its c.v.

References

Perron, Pierre, and Tomoyoshi Yabu. “Testing for Shifts in Trend With an Integrated or Stationary Noise Component.” Journal of Business & Economic Statistics 27, no. 3 (July 2009): 369–96. https://doi.org/10.1198/jbes.2009.07268.

uroot.ADF *A simple implementation of ADF test*

Description

A function for ADF test with the ability to select the number of lags. Lags are selected by informational criterions which can be modified as in Ng and Perron (2001) and Cavaliere et al. (2015).

Usage

```
uroot.ADF(
  y,
  const = TRUE,
  trend = FALSE,
  max.lag = 0,
  criterion = NULL,
  modified.criterion = FALSE,
  rescale.criterion = FALSE,
  recursive = FALSE,
  cc = 0,
  gamma = 0,
  trim = 0.15,
  boot.p = FALSE,
  boot.iter = 999
)
```

Arguments

`y`	A time series of interest.
`const, trend`	Whether a constand and trend are to be included.
`max.lag`	Maximum lag number.
`criterion`	A criterion used to select number of lags. If lag selection is not needed keep this NULL.
`modified.criterion`	Whether the unit-root test modificaton is needed.
`rescale.criterion`	Whether the rescaling informational criterion is needed. Designed to cope with heteroscedasticity in residuals.
`recursive`	Whether a recursive detrending should be applied. See Smeekes (2013) for details.
`cc`	A filtration parameter used to construct an autocorrelation coefficient.
`gamma`	Detrending type selection parameter. If 0 the OLS detrending is applied, if 1 the GLS detrending is applied, otherwise the autocorrelation coefficient is calculated as $1 + c^{\gamma}T^{-\gamma}$.
`trim`	A trimming parameter.
`boot.p`	Whether bootstrapped p-values should be returned.
`boot.iter`	The number of bootstrap iterations.

Details

Due to the Frisch-Waugh-Lovell theorem we first detrend `y` and then apply the test to the detrended series.

Value

A list containing:

- y,
- const,
- trend,
- residuals,
- coefficient estimates,
- t-statistic value,
- critical value,
- Number of lags,
- indicator of stationarity.

References

Cavaliere, Giuseppe, Peter C. B. Phillips, Stephan Smeekes, and A. M. Robert Taylor. “Lag Length Selection for Unit Root Tests in the Presence of Nonstationary Volatility.” Econometric Reviews 34, no. 4 (April 21, 2015): 512–36. https://doi.org/10.1080/07474938.2013.808065.

Ng, Serena, and Pierre Perron. “Lag Length Selection and the Construction of Unit Root Tests with Good Size and Power.” Econometrica 69, no. 6 (2001): 1519–54. https://doi.org/10.1111/1468-0262.00256.

Taylor, A. M. Robert. “Regression-Based Unit Root Tests With Recursive Mean Adjustment for Seasonal and Nonseasonal Time Series.” Journal of Business & Economic Statistics 20, no. 2 (April 2002): 269–81. https://doi.org/10.1198/073500102317352001.

MacKinnon, James G. “Critical Values for Cointegration Tests.” Working Paper. Economics Department, Queen’s University, January 2010. https://ideas.repec.org/p/qed/wpaper/1227.html.

Smeekes, Stephan. “Detrending Bootstrap Unit Root Tests.” Econometric Reviews 32, no. 8 (July 2013): 869–91. https://doi.org/10.1080/07474938.2012.690693.

Elliott, Graham, Thomas J. Rothenberg, and James H. Stock. “Efficient Tests for an Autoregressive Unit Root.” Econometrica 64, no. 4 (1996): 813–36. https://doi.org/10.2307/2171846.

`uroot.CHLT` *MDF test for a single break and possible heteroscedasticity*

Description

MDF test for a single break and possible heteroscedasticity

Usage

```
uroot.CHLT(y, max.lag = NULL, trim = 0.15, boot.cv = FALSE, boot.iter = 999)
```

Arguments

`y`	A time series of interest.
`max.lag`	The maximum possible lag.
`trim`	Trimming parameter for lag selection
`boot.cv`	Whether we need bootstrapped critical values.
`boot.iter`	Number of bootstrap iterations.

Details

The code provided is the original GAUSS code by Cavaliere et al. ported to R.

Value

An object of type `mdfCHLT`. It's a list of four sublists each containing:

- The value of MZ_α, MSB, MZ_t, or ADF,
- The asymptotic c.v.,
- The bootstrapped c.v.

References

Cavaliere, Giuseppe, David I. Harvey, Stephen J. Leybourne, and A.M. Robert Taylor. "Testing for Unit Roots in the Presence of a Possible Break in Trend and Nonstationary Volatility." Econometric Theory 27, no. 5 (October 2011): 957–91. https://doi.org/10.1017/S0266466610000605.

`uroot.HLT` *Unit root testing procedure under a single structural break.*

Description

Unit root testing procedure under a single structural break.

Usage

```
uroot.HLT(y, const = FALSE, trim = 0.15)
```

Arguments

`y`	A time series of interest.
`const`	Whether a constant should be included.
`trim`	The trimming parameter to find the lower and upper bounds of possible break dates.

Value

The value of test statistic.

References

Harvey, David I., Stephen J. Leybourne, and A. M. Robert Taylor. "Unit Root Testing under a Local Break in Trend." Journal of Econometrics 167, no. 1 (2012): 140–67.

`uroot.robust` — *A wrapping function around MDF.1br.*

Description

A wrapping function around MDF.1br.

Usage

```
uroot.robust(
  y,
  const = FALSE,
  trend = FALSE,
  season = FALSE,
  trim = 0.15,
  criterion = "bic"
)
```

Arguments

`y`	A time series of interest.
`const, trend`	Whether the constant term and trend should be included.
`season`	Whether the seasonal adjustment is needed.
`trim`	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

`uroot.robust.mlt` — *A wrapping function around breaktest.KP and MDF.mlt.*

Description

A wrapping function around breaktest.KP and MDF.mlt.

Usage

```
uroot.robust.mlt(y, const = FALSE, season = FALSE, breaks = 2, trim = 0.15)
```

Arguments

`y`	A series of interest.
`const`	Whether the constant term should be included.
`season`	Whether the seasonal adjustment is needed.
`breaks`	Number of breaks.
`trim`	Trimming value for a possible break date bounds.

Details

The code provided is the original Ox code by Skrobotov (2018) ported to R.

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uroot.SADF *Supremum ADF tests*

Description

Supremum ADF tests

Usage

```
uroot.SADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  add.p.value = TRUE,
  boot.p = FALSE,
  boot.iter = 999
)

uroot.GSADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  add.p.value = TRUE,
  boot.p = FALSE,
  boot.iter = 999
)
```

Arguments

`y`	A time series of interest.
`trim`	A trimming parameter to determine the lower and upper bounds for a possible break point.
`const`	Whether the constant needs to be included.
`add.p.value`	Whether the p-value is to be returned. This argument is needed to suppress the calculation of p-values during the precalculation of tables needed for the p-values estimating.

Details

`SADF.test` is a test statistic equal to the minimum value of uroot.ADF for subsamples starting at $t = 1$.

`GSADF.test` is a generalized version of `SADF.test`. Subsamples are allowed to start at any point between 1 and $T(1 - trim)$.

Value

An object of type `sadf`. It's a list of:

- `y`,
- `trim`,
- `const`,
- vector of t-values,
- the value of the corresponding test statistic,
- p-value if it was asked for.

References

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." Journal of Financial Econometrics, April 23, 2022. https://doi.org/10.1093/jjfinec/nbac004.

`uroot.sb.GSADF` *Sign-based SADF test*

Description

Sign-based SADF test

Usage

```
uroot.sb.GSADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 999,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)
```

Arguments

`y`	A time series of interest.
`trim`	A trimming parameter to determine the lower and upper bounds for a possible break point.
`const`	Whether the constant needs to be included.
`alpha`	Needed level of significance.
`iter`	Number of bootstrapping iterations.
`urs`	Use union of rejections strategy if `TRUE`.
`seed`	The seed parameter for the random number generator.

Details

Refactored original code by Kurozumi et al.

References

Harvey, David I., Stephen J. Leybourne, and Yang Zu. “Sign-Based Unit Root Tests for Explosive Financial Bubbles in the Presence of Deterministically Time-Varying Volatility.” Econometric Theory 36, no. 1 (February 2020): 122–69. https://doi.org/10.1017/S0266466619000057.

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. “Time-Transformed Test for Bubbles under Non-Stationary Volatility.” Journal of Financial Econometrics, April 23, 2022. https://doi.org/10.1093/jjfinec/nbac004.

`uroot.STADF` *Supremum ADF tests with time transformation*

Description

Supremum ADF tests with time transformation

Usage

```
uroot.STADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = FALSE,
  omega.est = TRUE,
  truncated = TRUE,
  is.reindex = TRUE,
  ksi.input = "auto",
  hc = 1,
  pc = 1,
  add.p.value = TRUE
)

uroot.GSTADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = FALSE,
  omega.est = TRUE,
  truncated = TRUE,
  is.reindex = TRUE,
  ksi.input = "auto",
  hc = 1,
  pc = 1,
  add.p.value = TRUE
)
```

Arguments

`y`	A time series of interest.
`trim`	A trimming parameter to determine the lower and upper bounds for a possible break point.
`const`	Whether the constant needs to be included.
`omega.est`	Whether the variance of Nadaraya-Watson residuals should be used.

`truncated`	Whether the truncation of Nadaraya-Watson residuals is needed.
`is.reindex`	Whether the Cavaliere and Taylor (2008) time transformation is needed.
`ksi.input`	The value of the truncation parameter. Can be either `auto` or the explicit numerical value. In the former case the numeric value is estimated.
`hc`	The scaling parameter for Nadaraya-Watson bandwidth.
`pc`	The scaling parameter for the estimated truncation parameter value.
`add.p.value`	Whether the p-value is to be returned. This argument is needed to suppress the calculation of p-values during the precalculation of tables needed for the p-values estimating.

Details

See uroot.SADF. Tests with time transformation are the modified versions of the ordinary `SADF` and `GSADF` tests using Nadaraya-Watson residuals and reindexing procedure by Cavaliere-Taylor (2008).

Value

An object of type `sadf`. It's a list of:

- `y`,
- `N`: Number of observations,
- `trim`,
- `const`,
- `omega.est`,
- `truncated`,
- `is.reindex`,
- `new.index`: the vector of new indices,
- `ksi.input`,
- `hc`,
- `h.est`,
- `u.hat`,
- `pc`,
- `w.sq`,
- `t.values`: vector of t-values,
- the value of the corresponding test statistic,
- `u.hat.truncated`: truncated residuals if truncation was asked for,
- `ksi`, `sigma`: estimated values of the truncation parameter and resulting s.e. if `ksi.input` equals `auto`,
- `eta.hat`: the values of reindexing function if reindexing was asked for,
- p-value if it was asked for.

References

Cavaliere, Giuseppe, and A. M. Robert Taylor. "Time-Transformed Unit Root Tests for Models with Non-Stationary Volatility." Journal of Time Series Analysis 29, no. 2 (March 2008): 300–330. https://doi.org/10.1111/j.1467-9892.2007.00557.x.

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." Journal of Financial Econometrics, April 23, 2022. https://doi.org/10.1093/jjfinec/nbac004.

`uroot.w.SADF` *Weighted supremum ADF test*

Description

Weighted supremum ADF test

Usage

```
uroot.w.SADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 4 * 200,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)

uroot.w.GSADF(
  y,
  trim = 0.01 + 1.8/sqrt(length(y)),
  const = TRUE,
  alpha = 0.05,
  iter = 4 * 200,
  urs = TRUE,
  seed = round(10^4 * sd(y))
)
```

Arguments

`y`	A time series of interest.
`trim`	The trimming parameter to find the lower and upper bounds of possible break dates.
`const`	Whether the constant needs to be included.
`alpha`	A significance level of interest.
`iter`	Nnumber of iterations.
`urs`	Use `union of rejections` strategy.
`seed`	A seed parameter for the random number generator.

Value

An object of type `sadf`. It's a list of:

- `y`,
- `trim`,
- `const`,
- `alpha`,

- `iter`,
- `urs`,
- `seed`,
- `sigma.sq`: the estimated variance,
- `BZ.values`: a series of BZ-statistic,
- `supBZ.value`: the maximum of `supBZ.values`,
- `supBZ.bootstsrap.values`: bootstrapped supremum BZ values,
- `supBZ.cr.value`: supremum BZ α critical value,
- `p.value`,
- `is.explosive`: 1 if `supBZ.value` is greater than `supBZ.cr.value`.

if `urs` is `TRUE` the following items are also included:

- vector of t-values,
- the value of the SADF test statistic,
- `SADF.bootstrap.values`: bootstrapped SADF values,
- `U.value`: union test statistic value,
- `U.bootstrap.values`: bootstrapped series of `U.value`,
- `U.cr.value`: critical value of `U.value`.

References

Harvey, David I., Stephen J. Leybourne, and Yang Zu. "Testing Explosive Bubbles with Time-Varying Volatility." Econometric Reviews 38, no. 10 (November 26, 2019): 1131–51. https://doi.org/10.1080/07474938.2

Kurozumi, Eiji, Anton Skrobotov, and Alexey Tsarev. "Time-Transformed Test for Bubbles under Non-Stationary Volatility." Journal of Financial Econometrics, April 23, 2022. https://doi.org/10.1093/jjfinec/nbac004.

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